Exp No: 2	Study of Network cables - straight-through and crossover cable

AIM:

To study the different types of network cables and to make a straight-through cable and a crossover cable using an RJ45 connector and a crimping tool.

APPARATUS REQUIRED:

- Ethernet cable (CAT5 / CAT5e)
- RJ45 connectors
- Crimping tool
- Cable tester (if available)

ALGORITHM / PROCEDURE:

1. Take an Ethernet cable and cut it to the required length.
2. Strip about 1.5 cm of the outer covering of the cable.
3. Untwist and separate the four pairs of wires.
4. Arrange the wires according to the required wiring standard.
5. For a **straight-through cable**, use the same arrangement at both ends.
6. For a **crossover cable**, use arrangement A at one end and arrangement B at the other end.
7. Cut the wires evenly and insert them into the RJ45 connector.
8. Make sure that all the wires reach the end of the connector in the correct order.
9. Insert the RJ45 connector into the crimping tool and press firmly.
10. Repeat the process for the other end of the cable.
11. Test the completed cable by connecting it to the appropriate devices.

Types of network cables:

Cable	Full form / description
UTP	Unshielded Twisted Pair
STP	Shielded Twisted Pair
Coaxial cable	Copper core with insulation and shield, used in cable TV and old LANs
Fibre optic cable	Glass / plastic fibre carrying light, used for long distance and backbone links

DIAGRAM:

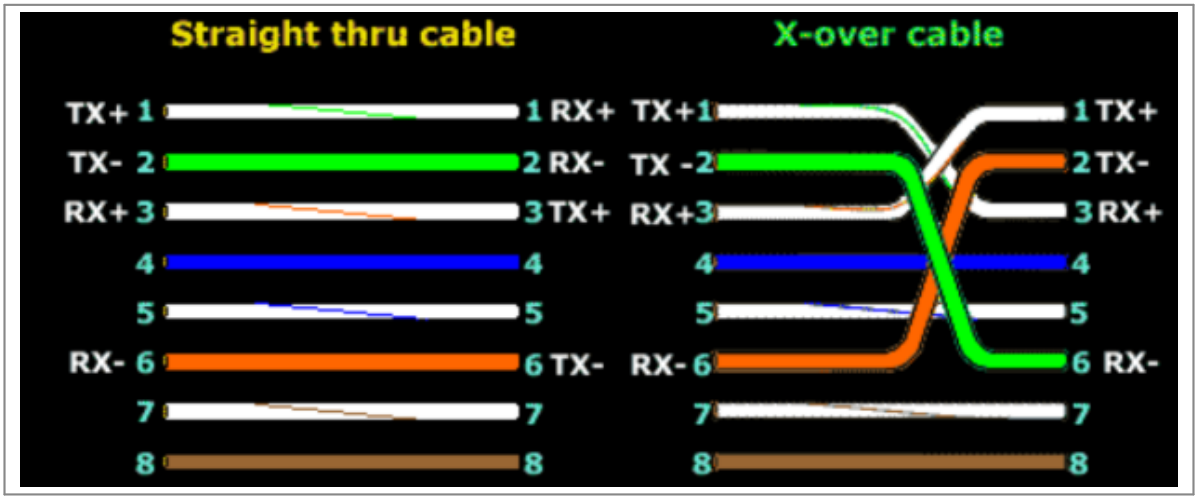


Fig 1: Straight-through cable and crossover (X-over) cable pin connections

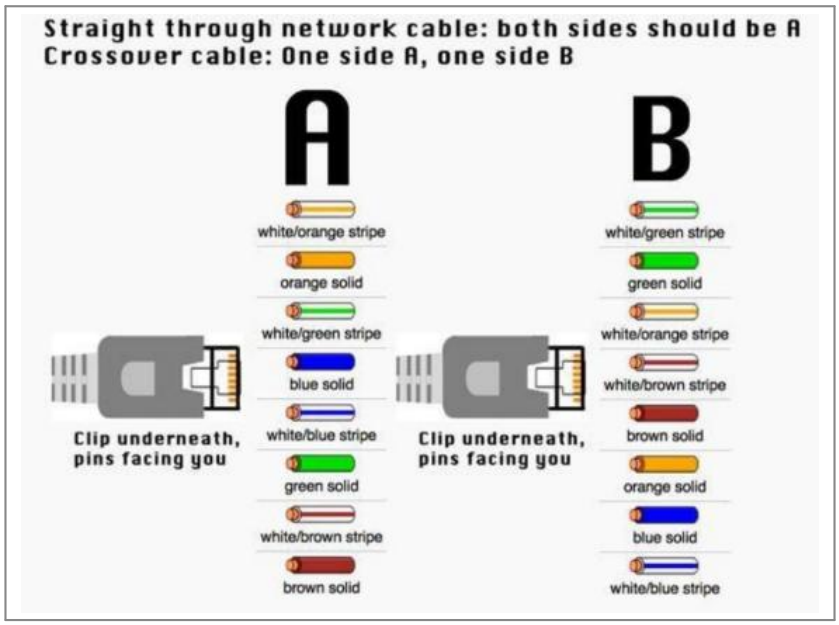


Fig 2: Wire arrangements A and B (straight-through = same arrangement at both ends, crossover = A at one end, B at the other)

OUTPUT:

Observation

Cable	Wiring arrangement	Usage
Straight-through cable	Same standard at both ends	PC to Switch / Router
Crossover cable	Different standards at each end	PC to PC

RESULT:

Thus the different types of network cables were studied, and straight-through and crossover Ethernet cables were prepared using RJ45 connectors and a crimping tool. The cables were tested and found to be working properly.